Amazon India Sales Performance Analysis
March – June 2022

Background

This analysis was conducted on behalf of an Amazon India marketplace seller operating across multiple Indian states. The business sells fashion and apparel products ,primarily Sets, Kurtas, Western Dress, and Tops fulfilled through both Amazon and Merchant channels. The dataset covers transactional sales data from March to June 2023 and was sourced from Kaggle for portfolio demonstration purposes.
The analytical brief was to identify primary revenue drivers, evaluate cancellation behaviour across geographic, product, size, and operational dimensions, and provide targeted recommendations to improve business performance.

Dataset Source: Kaggle Amazon Sale Report (India, 2022)

North Star Metric

Confirmed Revenue Rate: 93%
The primary measure of business health for this analysis is the proportion of total order value that converts into confirmed revenue. A confirmed revenue rate of 93% indicates that the business retains the large majority of order value, with cancellations representing a contained and manageable 7% of total order value. Maintaining and improving this rate is the central objective driving all recommendations in this report.

1. Executive Summary

This analysis examines sales performance across confirmed and cancelled orders for an Amazon India marketplace dataset covering March–June 2022. The objective was to identify primary revenue drivers, evaluate cancellation behaviour across multiple dimensions, and provide actionable recommendations to improve business performance.

Dataset Note: Analysis was conducted on three months of recent transactional data. While the time window limits seasonality conclusions, recent data reflects current consumer behaviour and operational conditions more accurately than older historical datasets.

Key Findings:

- 93% of total order value comes from confirmed orders, while 7% comes from cancellations
- Revenue is highly concentrated: the top five states contribute 56.41% of confirmed revenue, with Maharashtra alone accounting for 17.11%
- Four categories dominate revenue generation: Set (47.58%), Kurta (28.97%), Western Dress (14.88%), and Top (6.40%), collectively accounting for over 97% of confirmed revenue

- State × Category analysis reveals cancellation behaviour largely mirrors the underlying sales distribution, with only minor localised deviations in Karnataka and Telangana show modest elevated cancellation concentration in Set (−3.09% and −2.37% respectively)
- Category × Size analysis identifies more meaningful variation than geographic analysis. Ethnic Dress (L size, −14.59%), Blouse (M size, −7.77%), and Bottom (XL size, −6.95%) show the strongest cancellation pressure, suggesting product-level factors are a stronger cancellation driver than geography
- Fulfilment channel analysis shows no systemic risk. Amazon and Merchant maintain broadly consistent confirmed and cancelled distributions across all major categories.
- Promotions are dominated by a single Free Shipping campaign (45.26% of confirmed orders), while 72.15% of cancelled orders have no promotion IDs, limiting causal analysis
- Revenue showed short-term volatility between 631K and 1.1M across the analysis period, with no single directional trend identified

Overall Assessment:

The business shows strong, stable performance overall, with a 93% confirmed order rate and no systemic cancellation issues across geography, fulfilment channel, or category. Improvement opportunities are targeted and actionable, mainly concentrated in specific product-size combinations rather than broad operational weaknesses. This allows the business to focus improvement efforts precisely instead of applying broad changes across all operations.

2. Business Logic Definition

2.1 Confirmed Orders (Revenue-Contributing)

Includes: Pending, Pending – Waiting for Pickup, Shipped (all successful delivery stages)
Assumption: Pending orders are treated as paid transactions awaiting fulfilment.

2.2 Cancelled Orders (Non-Revenue)

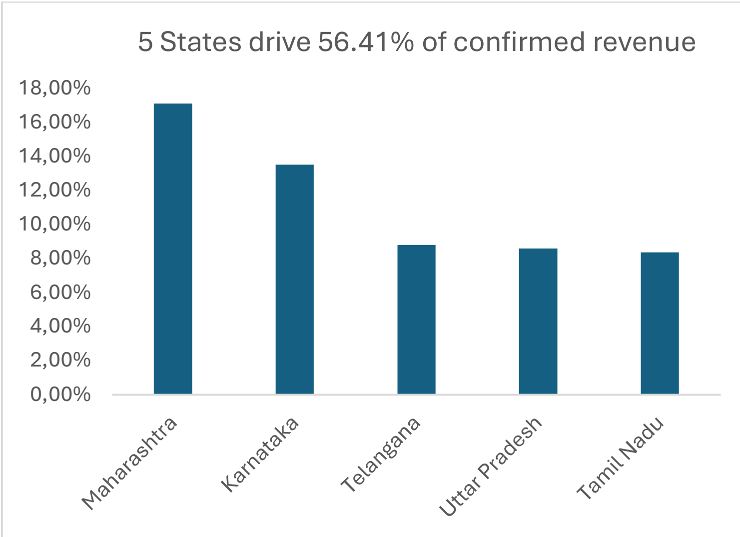
Includes: Cancelled, Returned/Returning to Seller, Damaged/Lost/Rejected shipments
These are excluded from revenue calculations due to refund or failure status.

3. Overall Performance

Metric	Value
Total Order Value	₹76,034,406
Confirmed Orders	₹70,847,184 (93%)
Cancelled Orders	₹5,187,222 (7%)

Insight: The business shows a strong conversion rate with cancellations remaining a relatively low and stable proportion of total order value.

4. Geographic Revenue Distribution



Across remaining states: 14 states contribute 38.87% collectively, while 25 states each contribute less than 4.72% individually, averaging only 0.19% each.

Insight: Revenue is heavily concentrated in a small number of high-performing states.

Recommendation: Business performance is dependent on a limited geographic base, making these five regions strategically critical for inventory planning, marketing investment, and customer retention.

5. State-Level Cancellation Pressure Analysis

Telangana and Uttar Pradesh Show Elevated Cancellation Pressure

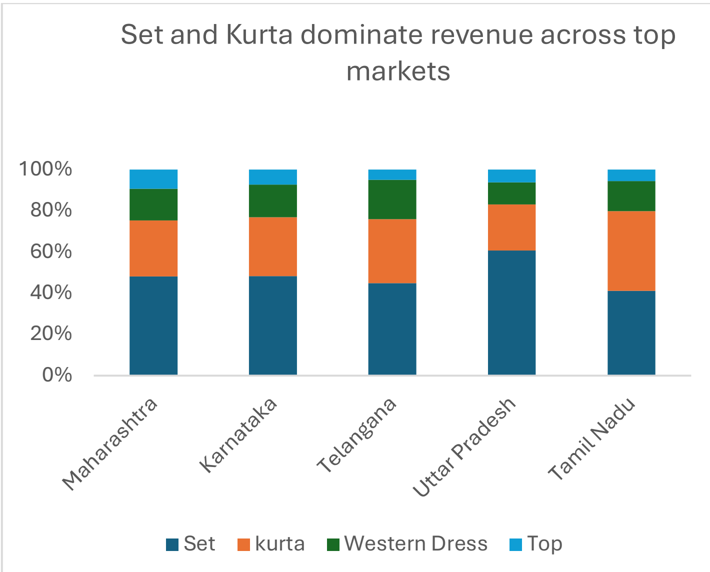
Metric: Cancellation Pressure Score (CPS) = Confirmed % − Cancelled %

State	CPS
Maharashtra	1,34%
Karnataka	1,26%
Telangana	-0,28%
Uttar Pradesh	-0,37%
Tamil Nadu	0,81%

A positive CPS indicates a state contributes proportionally more to confirmed orders than cancelled orders. A negative CPS indicates the state is disproportionately represented in cancellations relative to its confirmed revenue share. Values close to 0% indicate proportional, neutral performance. The more negative the value, the greater the relative cancellation concentration.

Insight: Cancellation behaviour closely tracks sales distribution across all five states. Telangana and Uttar Pradesh show slightly higher relative cancellation concentration, but deviations are minimal and fall within normal variation. No major state-level cancellation risk is identified.

6. Category Performance Across Top States



Ethnic Dress, Blouse, Bottom, Saree, and Dupatta together contribute less than 2% on average across the top five states.

Insight: Four categories account for over 97% of confirmed revenue generation across the highest-performing geographic markets.

Recommendation: Inventory planning, forecasting, and promotional investment should prioritise Set, Kurta, Western Dress, and Top.

7. State × Category Diagnostic Analysis

Cancellation Pressure Is Localised Across State-Category Combinations
Metric: CPS = Confirmed % − Cancelled %

Category	Maharashtra	Karnataka	Telangana	Uttar Pradesh	Tamil Nadu
Set	-0,43%	-3,09%	-2,37%	-1,63%	-1,44%
Kurta	-1,32%	-1,08%	2,87%	0,79%	-0,77%
Western Dress	2,40%	2,27%	1,41%	0,21%	-0,30%
Top	-1,35%	1,40%	-1,78%	-0,24%	1,33%
Ethnic Dress	0,32%	0,07%	-0,58%	0,52%	0,64%

Positive values indicate a category contributes proportionally more to confirmed orders than cancelled orders within a state. Negative values indicate the category represents a larger share of cancelled orders than confirmed orders.

Materiality Framework:

- ±0–2%: Normal variation, no action required
- ±2–4%: Worth monitoring, particularly for high-volume categories
- ±5% and above: Meaningful business signal requiring investigation

Blouse, Bottom, Saree, and Dupatta were excluded from interpretation due to very low contribution to overall revenue and cancellations, making reliable patterns difficult to establish.

Key Insights:

Stable — Most category distributions remain consistent. Across the five highest-revenue states, most differences fall within the ±2% normal variation band, indicating cancellation behaviour generally follows the underlying sales distribution. No widespread state-level category problems are identified.

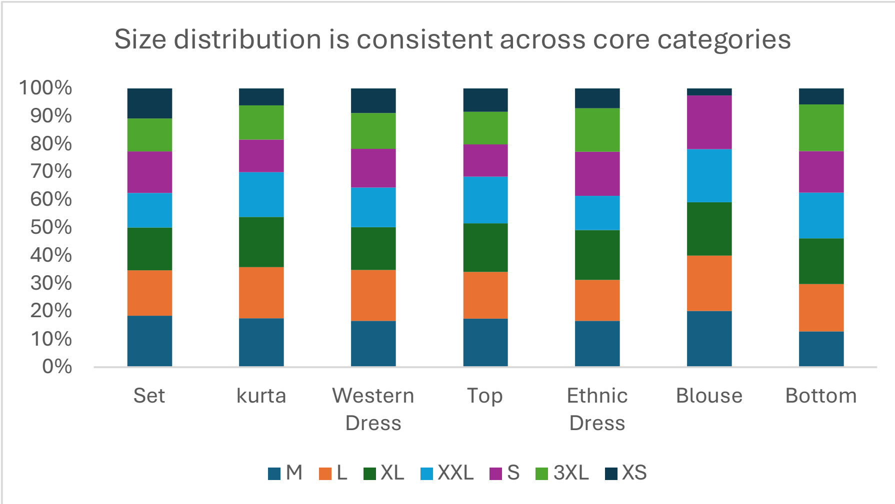
Notable — Set shows moderate cancellation concentration in Karnataka and Telangana. Karnataka (−3.09%) and Telangana (−2.37%) both fall within the monitor band. Although the deviations are modest, Set is the highest-volume category, making even small shifts operationally meaningful.

Stable — Kurta performs particularly well in Telangana. At +2.87%, Kurta represents a proportionally larger share of confirmed than cancelled orders in this market, indicating a relative strength.

Stable — Maharashtra and Tamil Nadu remain well balanced. Both states display only minor differences across all principal revenue-generating categories, with no deviation approaching the monitoring threshold.

Business Conclusion: The State × Category analysis indicates that geographic performance is generally stable, with cancellations largely reflecting the underlying sales mix. Only a small number of localised deviations stand out. Set in Karnataka and Telangana warrants monitoring given high category volume, while all other combinations remain within normal variation. Cancellation risk appears driven by specific localised factors rather than broad geographic trends.

8. Category × Size Diagnostic Analysis



Ethnic Dress (L) and Blouse (M) Carry the Highest Cancellation Pressure
Metric: CPS = Confirmed % − Cancelled %

Size	Set	Kurta	Western Dress	Top	Ethnic Dress	Blouse	Bottom
M	-1,86%	-1,00%	-0,39%	-2,07%	0,84%	-7,77%	-1,12%
L	-0,16%	-0,22%	0,85%	-0,61%	-14,59%	7,90%	3,22%
XL	0,29%	-0,28%	0,52%	-0,77%	7,98%	-1,19%	-6,95%
XXL	1,43%	0,35%	0,77%	3,03%	-2,68%	8,53%	5,23%
S	0,17%	-1,13%	-3,49%	-0,67%	1,03%	-4,72%	-0,35%
3XL	1,05%	1,27%	2,32%	2,13%	3,58%	—	4,49%
XS	-1,12%	-0,76%	-0,56%	-1,06%	3,83%	-0,42%	-4,53%

4XL, 5XL, 6XL, Free Size, Saree, and Dupatta were excluded due to very low order volumes insufficient to establish meaningful patterns.

Key Insights:

Notable — Ethnic Dress shows the strongest single deviation in the analysis. L-sized Ethnic Dresses are heavily overrepresented in cancelled orders (−14.59%), while XL sizes perform substantially better (+7.98%). This is the clearest indication that a specific product-size combination warrants further operational investigation, whether into sizing accuracy, product description, or customer expectations.

Notable — Blouse shows the largest range of variation across sizes. M-sized Blouses show strong cancellation concentration (−7.77%), while L (+7.90%) and XXL (+8.53%) perform considerably better. This suggests cancellation behaviour differs considerably across Blouse sizes and may point to sizing inconsistencies or fit issues at the M size.

Moderate — Bottom shows noticeable variation across sizes. XL (−6.95%) and XS (−4.53%) show elevated cancellation concentration, while L (+3.22%) and XXL (+5.23%)

perform positively. Although Bottom contributes a relatively small share of total revenue, the variation is consistent enough to warrant monitoring.

Stable — Set and Kurta remain remarkably consistent. Despite being the highest-volume categories, most deviations remain within ±2%, indicating cancellations closely follow their underlying sales distribution across nearly every size.

Stable — Western Dress and Top show only modest variation. The largest observed deviation is Western Dress at S size (−3.49%), which remains small relative to the significant anomalies identified in Blouse and Ethnic Dress.

Business Conclusion: Unlike the State × Category analysis, which found cancellation behaviour to be largely stable across geographic markets, the Category × Size analysis reveals more meaningful localised variation. This suggests product-level factors are a stronger driver of cancellation risk than geography. The most actionable findings are concentrated in Ethnic Dress (L size), Blouse (M size), and Bottom (XL and XS) — specific enough that a merchandising or operations team could act on them immediately. Set and Kurta, the two highest-revenue categories, maintain stable size distributions, suggesting consistent performance across their primary size ranges.

9. Fulfilment Channel Analysis

Fulfilment Channel Shows Minimal Cancellation Disparity Overall
Metric: CPS = Confirmed % − Cancelled %

Overall Fulfilment Performance

Channel	Confirmed	Cancelled	CPS
Amazon	71,92%	72,50%	−0,58%
Merchant	28,08%	27,50%	+0,58%

Overall Insight: The fulfilment split is almost identical across confirmed and cancelled orders. There is no evidence that one fulfilment channel contributes disproportionately to cancellations overall. This is a positive finding, fulfilment channel by itself does not appear to be a major driver of cancellations.

10. Promotion Analysis

Confirmed Orders

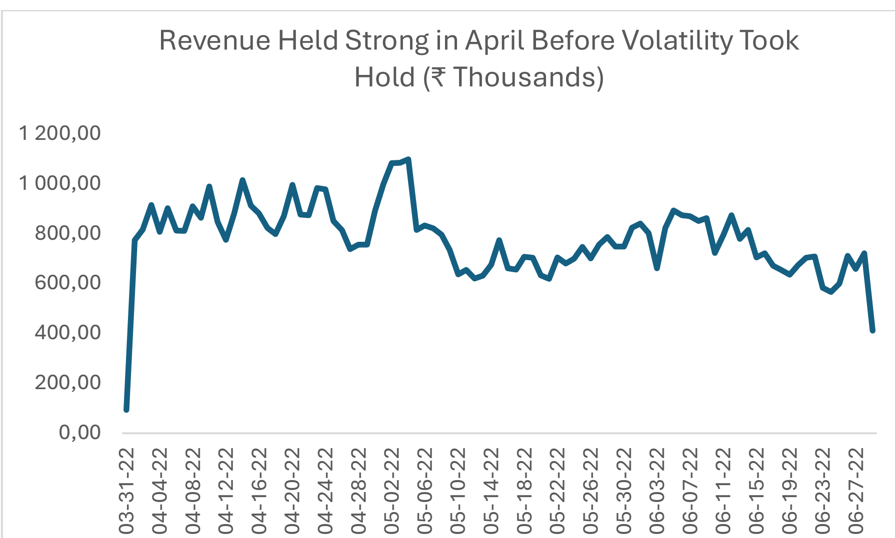
Promotion	Share
IN Core Free Shipping 2015/04/08	45.26%
No/Unknown Promotion ID	25.68%
All other promotions	29.06% (fragmented, averaging ~0.01% each)

Cancelled Orders

Promotion	Share
Unknown/No Promotion ID	72.15%

Insight: The Free Shipping promotion is the strongest single contributor to confirmed orders. However, because nearly three-quarters of cancelled orders carry an unknown promotion ID, it is not possible to reliably determine whether specific promotions influence cancellation behaviour.
Conclusion: No reliable evidence suggests promotions drive cancellations. Promotion analysis is limited by data quality rather than analytical methodology.

11. Time Analysis



The dataset covers March to June 2022. Due to the limited time window, this analysis observes short-term revenue behaviour rather than drawing conclusions about seasonality or long-term trends.

Confirmed order value opened the period in a stable range between 800K and 1.0M (31 March to 12 April), suggesting consistent baseline demand in early April. Following this consolidation phase, revenue became increasingly volatile. A notable low of 737K was recorded on 27 April, followed by the period's peak of 1.1M on 4 May. Performance then declined to 631K by 13 May before partially recovering to 893K by 5 June. The final data point of 413K on 29 June reflects an incomplete reporting period and should not be interpreted as a confirmed downward trend.

Insight: Revenue fluctuated across a range of approximately 631K to 1.1M during the analysis period. The volatility observed does not indicate a single directional trend but rather short-term instability that warrants monitoring over a longer horizon.

Recommendation: A longer historical dataset would be required to determine whether this fluctuation reflects seasonal demand patterns, promotional cycles, or operational factors. The category and geographic optimisations recommended below are expected to contribute to more stable revenue performance over time.

12. Key Business Recommendations

1. Geographic Focus
Prioritise Maharashtra, Karnataka, and Tamil Nadu for marketing, inventory planning, and customer retention initiatives given their high confirmed revenue concentration. Monitor Telangana and Uttar Pradesh where modest cancellation pressure exists, though neither represents a systemic risk.

2. Category Strategy
Focus inventory planning, forecasting, and promotional investment on Set, Kurta, Western Dress, and Top — the four categories collectively generating over 97% of confirmed revenue.

3. Operational Risk — Product Size
Conduct targeted investigations into the following specific product-size combinations, which showed the strongest cancellation pressure in the analysis:

- Ethnic Dress — L size (−14.59% CPS): the single largest deviation identified across the entire analysis
- Blouse — M size (−7.77% CPS): investigate sizing accuracy and fit consistency
- Bottom — XL and XS sizes (−6.95% and −4.53% respectively)

4. Fulfilment Channel
No major action required. Fulfilment channel was assessed as a priority area of investigation and found to carry minimal overall disparity between confirmed and cancelled orders.

5. Promotion Strategy
Scale Free Shipping-type promotions given their dominant contribution to confirmed orders (45.26%). Improve Promotion ID capture for cancelled orders to enable more robust future campaign performance analysis.

6. Data Quality Improvement
Improve Promotion ID assignment in cancelled orders. Only 27.85% of cancelled orders carry an identifiable promotion ID, severely limiting the ability to evaluate promotional influence on cancellation behaviour.

7. Time Series Extension
Collect a longer historical dataset to support seasonal demand analysis, promotional cycle evaluation, and revenue forecasting. The current three-month window, while operationally relevant, is insufficient for trend analysis.

13. Analytical Approach

This analysis followed a structured diagnostic flow:

- Business logic definition (Confirmed vs Cancelled order classification)
- Overall revenue and cancellation baseline establishment
- Geographic revenue concentration analysis
- Category contribution analysis across top markets
- Multi-dimensional CPS diagnostic analysis across state × category, category × size, and fulfilment channel dimensions
- Operational breakdown across fulfilment and promotion dimensions
- Short-term time series observation
- Business interpretation and prioritised recommendations

Cancellation Pressure Score (CPS) — a custom diagnostic metric developed for this analysis was applied consistently across all comparative dimensions. CPS = Confirmed % − Cancelled %, where values close to 0% indicate proportional, neutral performance; positive values indicate relative strength in confirmed orders; and negative values indicate disproportionate cancellation concentration. A materiality framework (±0–2% normal, ±2–4% monitor, ±5%+ investigate) was applied to distinguish meaningful signals from normal variation.

This project demonstrates end-to-end analytical capability including data cleaning, business rule definition, exploratory analysis, multi-dimensional diagnostic comparison, materiality-based prioritisation, and translation of data patterns into targeted business actions. Notably, the analysis correctly identifies both areas of risk and areas of stability recognising that confirming the absence of a problem is as analytically valuable as identifying one.